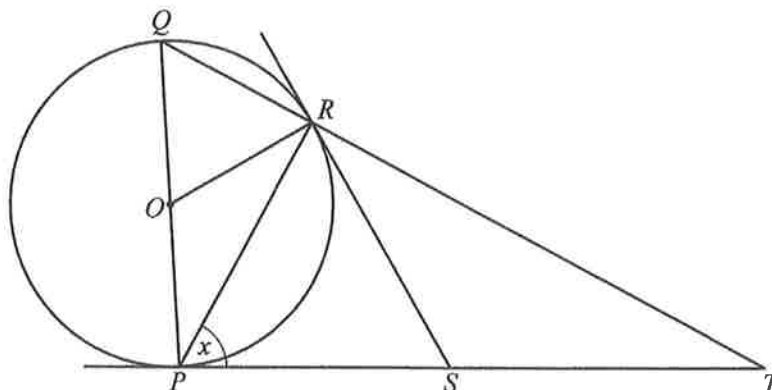


Intermediate

1.



The diagram shows a circle, centre O , with diameter PQ . The line PT is the tangent to the circle at P . The point R lies on the circle and the tangent to the circle at R meets PT at S . QRT is a straight line and angle $RPS = x$.

Prove, in either order, that

(i) angle $RST = 2x$

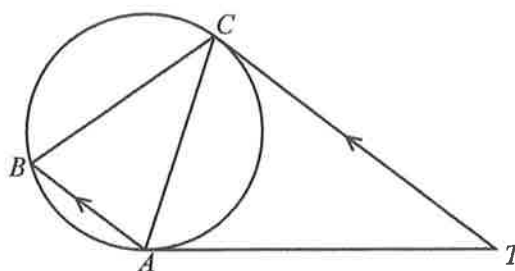
[3]

(ii) triangle RST is isosceles.

[3]

[N22/I/7]

2.



In the diagram, A , B and C lie on a circle. The tangents to the circle at A and C meet at T and BA is parallel to CT .

(a) Prove that triangle ABC is isosceles.

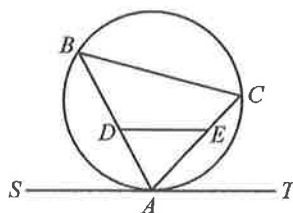
[4]

(b) Prove that angle $BCA = \text{angle } CTA$.

[3]

[N21/I/11]

3.

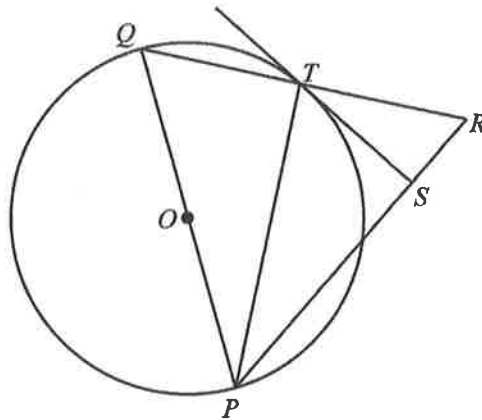


The diagram shows a circle passing through the points A , B and C . The straight line SAT is a tangent to the circle. The points D and E lie on AB and AC respectively and are such that $BCED$ is a cyclic quadrilateral. Prove that DE is parallel to ST .

[5]

[N20/II/4]

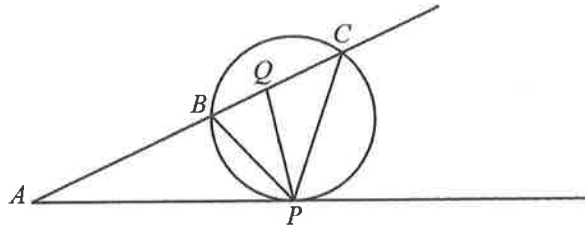
4.



In the diagram, PQ is the diameter of a circle, centre O . Triangle PQR is an isosceles triangle with $PQ = PR$. The line QR intersects the circle at T . The tangent to the circle at T meets PR at S .

- (i) Show that angle $TSR = 90^\circ$. [5]
 - (ii) Explain why the circle passing through the points S , R and T has its centre at the midpoint of TR . [2]
- [N19/I/6]

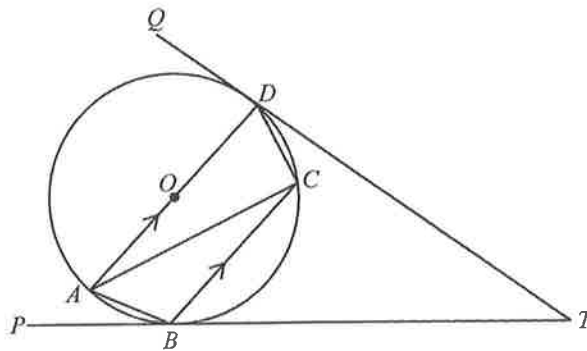
5.



In the diagram above, a line from a point A intersects a circle at the points B and C . The line AP is a tangent to the circle at the point P . The point Q lies on AC such that $AP = AQ$. Prove that PQ bisects angle BPC .

[N18/II/3]

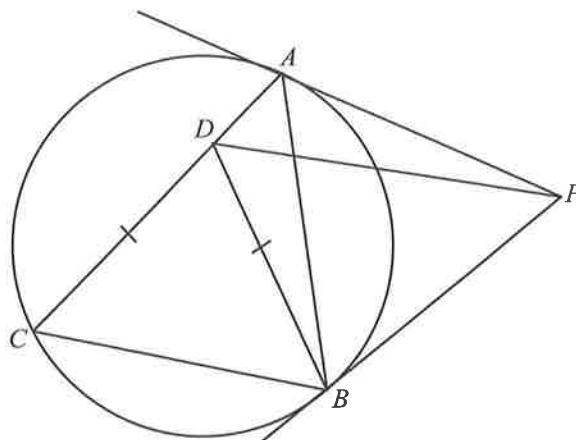
6.



The diagram shows a circle, centre O . Points A , B , C and D lie on the circle. The line AD is a diameter and BC is parallel to AD . The lines PBT and QDT are tangents to the circle at B and D respectively.

- (i) Show that angle $PBA = \text{angle } DAC$. [3]
 - (ii) Show that angle $CBT = 90^\circ - 2 \times (\text{angle } PBA)$. [5]
- [N17/I/10]

7.

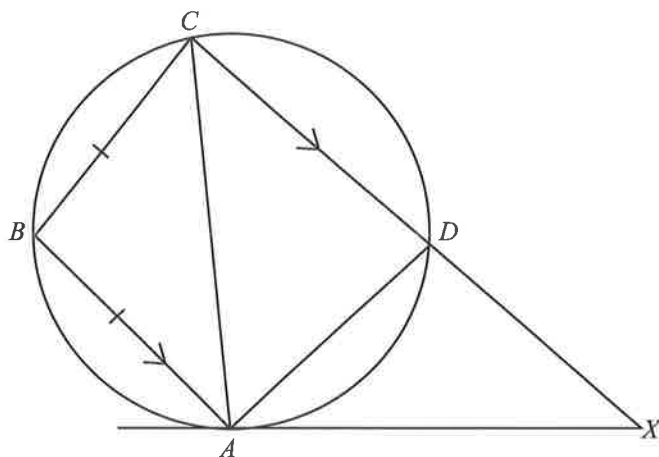


The diagram shows a circle passing through the vertices of a triangle ABC . The tangents to the circle at A and B intersect at the point P . The point D lies on AC such that $DC = DB$.

- (i) Prove that $\text{angle } APB + \text{angle } ADB = 180^\circ$. [5]
- (ii) Given that A, P, B and D lie on a circle, prove that PD and BC are parallel. [2]

[N16/II/5]

8.



In the diagram, $ABCD$ is a cyclic quadrilateral in which $BA = BC$ and BA is parallel to CD . The tangent to the circle at A meets CD produced at X .

- (i) Show that $\text{angle } DAX = \text{angle } BCA$. [4]
- (ii) Show that triangle ADX is isosceles. [3]

[N15/I/9]

